
STRESS
LESS

Moving forward

4 Information about stress

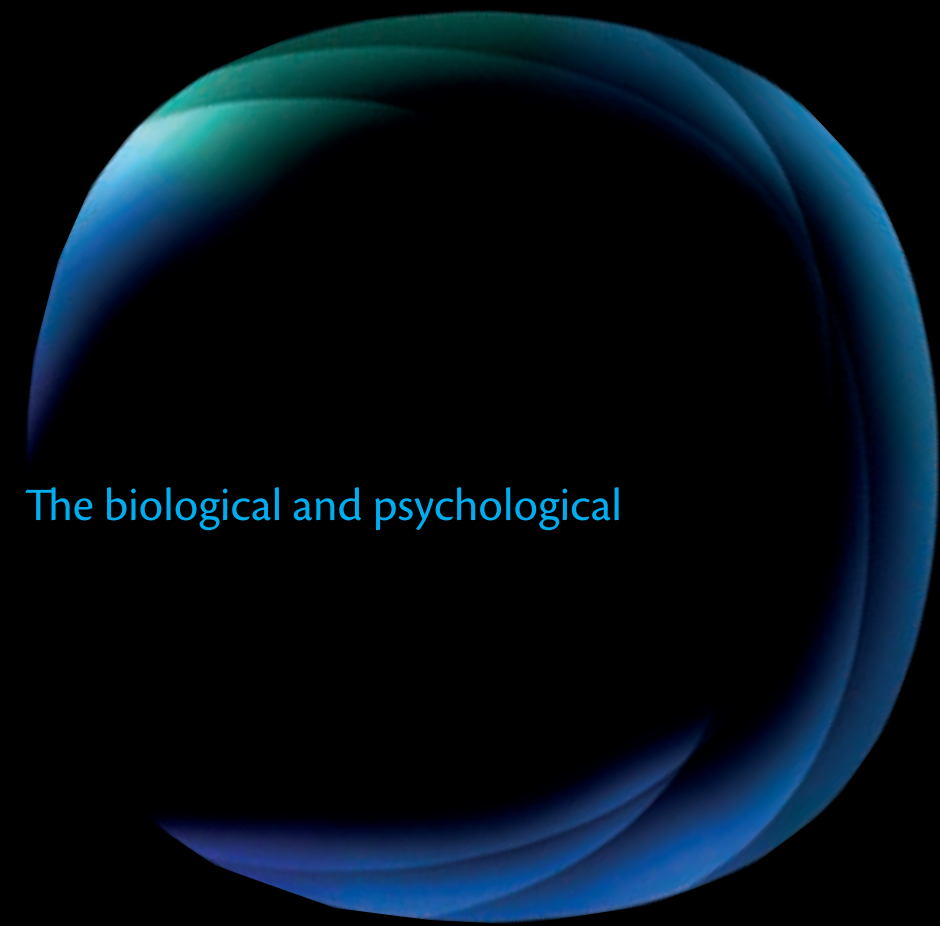
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Everything needed to know about stress



The biological and psychological

All of the work is based on research from documentaries, lectures, websites, and books cited at the end of the book.

A book by Alexander S. Park.

Many times we create

our own stress because of certain **negative perceptions**.

You can learn to correct them.

A huge component of stress is the lack of control and predictability of mostly our environment.

This means more psychological stress for most of those further down on a hierarchy.

Stress is difficult to define because it is so different for each of us. A good analogy is a rollercoaster. There are those in the back that view it as a torture-chamber, and those in the front who are thrill seekers, and those in the middle that are nonchalant.

What differentiated the passengers from the back to those up front was the **sense of control they had over the event and their environment**.

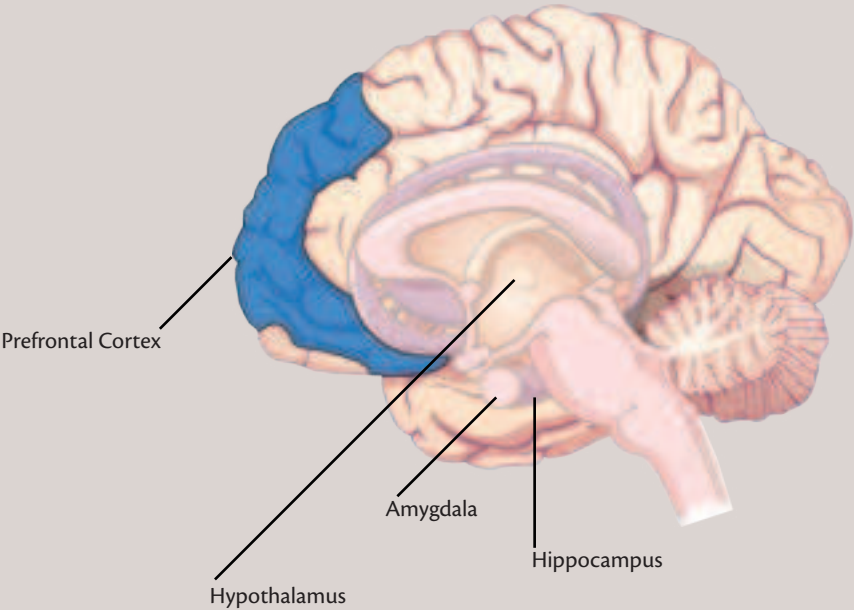
While neither group had any more or less control their perceptions and expectations were quite different.



People react to being on a rollercoaster differently.

When the brain detects a stressful situation, your body releases a hormone called Cortisol that primes your body for action. Chronic stress increases the activity level and number of neural connections in the Amygdala (the brain's fear center).

As levels of Cortisol rise, electric signals in your Hippocampus (learning, memory, stress control) deteriorate. The Hippocampus also inhibits the activity of the HPA Axis, and when this weakens so does the ability to control your stress.



Cortisol can cause the brain to shrink in size, resulting in the loss of synaptic connections between neurons and shrink the Prefrontal Cortex; **regulating behaviors like concentration, decision making, judgment, and social interaction**.

Fewer new brain cells are made in the Hippocampus, making it harder to learn and remember things, and set stage for more serious mental problems.

Scientists have known for years...

that elevated Cortisol levels:

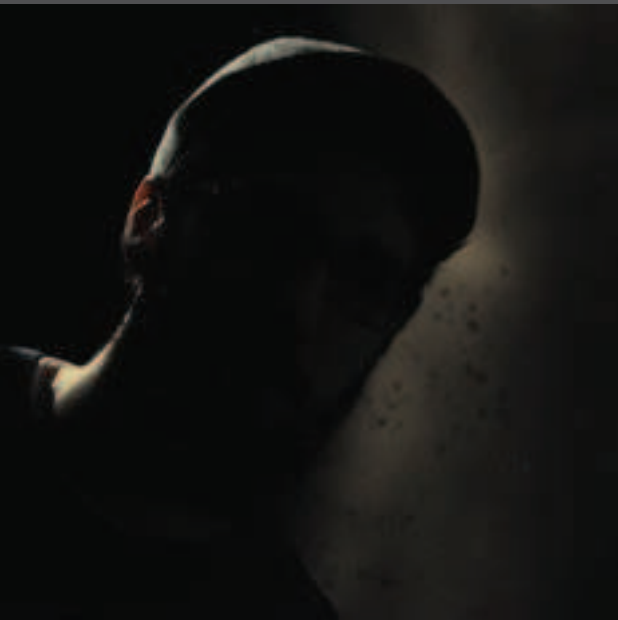
Interfere with learning and memory, lower immune function and bone density, increase weight gain, blood pressure, cholesterol, heart disease...

Cortisol disrupts the reproductive system, causing infertility or even miscarriage when Cortisol levels are too high or chronically elevated.

When our reproductive system shuts down, guess what hormone level goes down? Testosterone. This causes negative things to happen in regard to training and fitness goals.

Chronic stress and elevated Cortisol levels also increase risk for depression, mental illness, and lower life expectancy. Two separate studies were published inScience linking elevated Cortisol levels as a potential trigger for mental illness and decreased resilience—especially in adolescence.

Cortisol causes an increase in gastric acid production. When chronic, this can lead to reflux and other problems in the intestine. The decreased blood flow to the GI tract can also [cause incredible problems with digestion](#).



Cortisol is necessary but needs balance. If not your body will fail.



Too much Cortisol for too long will cause your body to burn out.

Increasing Cortisol and chronically elevated Cortisol can cause intense hunger and food cravings due to the metabolic derangement that occurs. If we let stress get the best of us for too long our bodies can't handle it anymore. Cortisol production may go through the roof, and then your body may not make enough...

[Without Cortisol, our bodies can not deal with the stress we encounter on an everyday basis.](#) However, when we abuse our minds and our bodies, we eventually pay the price. Our Cortisol levels become incongruent with our needs.

Stress affects the Hippocampus

(learning and memory)

Severe acute stress makes people stupid, making it impossible to remember in a very short period of time things you know perfectly well.

When there's less Dopamine in the brain, everything one would take pleasure in, is less pleasurable.

Serotonin regulates anxiety, happiness, and mood. Endorphins relieve pain, induce feelings of pleasure or euphoria.



Don't fall to acute stress, and take care of Dopamine levels in your brain.

Dopamine increases when we are organized and finish tasks – regardless if the task is small or large. When we're in creative mode, we can become hyper-focused. As a result, we can enter a state called flow. Dopamine is the brain chemical that allows us to achieve this state. The lesson is this: take up a hobby or activity in which you actually create something tangible.

Exercise increases multiple neurotransmitters – Serotonin and Endorphins, besides Dopamine, receive a boost. Simply taking a stroll or climbing some stairs will achieve a good Dopamine jolt. Tyrosine is considered the building block of Dopamine. There's a large list of foods that increase Tyrosine, including: Almonds, Avocados, Bananas, Beef, Chicken, Chocolate, Coffee, Eggs, Green Tea, Watermelon, Yogurt.

Listening to music can increase Dopamine levels temporarily, but what we're really looking for is a lasting fulfillment feeling so you can make your daily life enjoyable and productive for your goals. Also, popular music these days is often manufactured in such a way as to prey on your brain's chemical dependency, making much of music a form of substance addiction.

Meditation and mindfulness are essential to a healthy mind. Meditation can be a highly effective form of dopamine increase, [weeding out the mental influences which may be causing your chemistry to be less than desired](#). With the intent of reaching a state of self-fulfillment, it clears out mental clutter and replaces it with presence and fulfillment for being alive.

Curcumin: An active ingredient common in curry spices and turmeric.

L-theanine: Increases multiple neurotransmitters in the brain, including dopamine. Is in Green tea.

Ginkgo Biloba: Popular and believed to boost Dopamine levels and keep it circulating in the brain longer.

Acetyl-L-tyrosine: A healthy dose makes it easier for the brain to produce Dopamine.



The left and right parts of your brain is required for the creative spark.

Both left and right hemispheres of the brain are necessary for truly creative work. Which means there is no creativity center in the brain. This is because creative cognition is a complex mental phenomenon that involves multiple sequential acts that use widespread circuits throughout the brain.

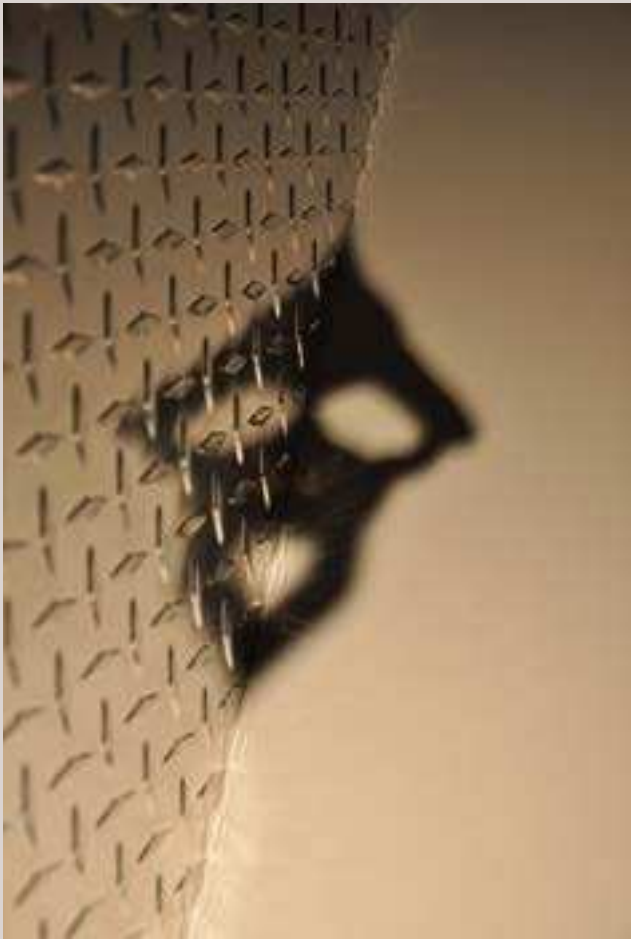
Both hemispheres of the brain can function independently. They are basically two complete brains; each with its own personality, likes, dislikes, and skills.

The left hemisphere does language, is logical & linear, while the right hemisphere is more synthetic; could do non-verbal reasonings, even some language functions.

Each hemisphere controls the opposite side of your body.

In response to the stress,

the body reacts using two different response mechanisms
the [central](#) stress response and the [peripheral](#) stress response.



It is possible for your stress to take control over you.

The central stress response is also known to decrease pain sensitivity and increase aggression or defensive behaviours.

The peripheral stress response causes an increase in glucose (for energy), circulation of oxygen and energy, increase in heart rate and blood pressure, and ultimately the activation of the fight or flight response.

non-essential functions are shut down such as, reproductive, inflammation, growth, and the immune system. – when that happens [stomach bacteria grows, and wipes out the bodies ability to heal itself.](#)

Multiple studies have found a link between creativity and Neuroticism — a tendency toward rumination and negative thinking. [Neuroticism is a personality trait that lends itself to worry, anxiety and isolation.](#)

Highly Neurotic people are more susceptible to mental illness than happy-go-lucky types. Studies have found, for example, that artists and other creative people score higher on tests of Neuroticism than people who aren't in creative fields. Neurotic people are imagining problems that don't exist.

Studies show that Neurotic people have a sensitive Amygdala, the almond-shaped brain structures involved in processing fear and anxiety. Neurotic people not only invent problems, but are very much stressed by them.



The face of Neuroticism.

Neurotic people also have an oversensitive Amygdala. [The Amygdala has a role in emotional responses, such as fear.](#) The Hypothalamus plays a role in emotional and behavioural responses, as well as part of the stress response (HPA axis). The tendency to become panicked over imagined problems can make Neurotic people quite miserable.

“If you dwell on problems for a long time... it seems quite obvious that you'll be more likely to come across a solution.”

Adam Perkins, a lecturer in the Neurobiology of personality at King's College London.

Negative affect is defined as a general dimension of subjective distress and unpleasurable engagement. High NA is characterized by irritability, hostility, nervousness, and shame, whereas low NA is characterized by calmness and serenity.

Found an association between negative affect and increased salivary Cortisol levels, as well as an association between positive affect and decreased Cortisol levels.



Positive affect (PA) indicates the degree to which a person feels active, enthusiastic, and pleasantly excited.



Negative affect (NA) is reflected by an inclination towards experiencing sadness and is a dimension of subjective distress.

Half of people with major depression have elevated levels of Glucocorticoids through the roof –something out of whack in regulation of this stress hormone during depression.

Statistically before the first major depressive episode, something awfully stressful occurs to a person and after coming out of it, they are no more at risk to depression than anyone else. Somewhere around the fourth or fifth stress-induced depression, things start cycling on their own, and they no longer need a major stressor to cause depression.

You feel more stressed, turn on a stress response, and are more at risk for a stress-related disease if there are no outlets for the frustration caused, no control over what’s happening, no predictability as to when its occurring, and nobodys shoulder to cry on.

This is falling into the cognitive psychologic soundbite of what a depression is. Depression is pathological extremes of feeling helpless, it is learned helplessness.

By circumstance where by any logic you should be saying this is awful but it’s not the whole world and do this cognitive distortion and decide this is indeed the whole world, and you have no control, always helpless, always hopeless.



Are you not enjoying a walk, nature, art, or beauty as you should?

Major stress can predispose you towards depression (whole lot of Glucocorticoids, but brain has depleted levels of Dopamine –the domain of anhedonia).

The absence of Dopamine has something to do with Anhedonia (inability to feel pleasure).

The absence of Norepinephrine has something to do with psychomotor retardation (hard to do and think stuff).

Absence of serotonin is this obsessive sense of grief.



Depression is considered a disorder on a biological level, it is not just a state of mind.

There is a gene that’s relevant to how readily we pick ourselves up after life has dumped us on our rear end, how readily we recover from stressors. Glucocorticoids (Cortisol) regulate the function of this gene. This is not something that’s “pull yourself together”, it’s as real of a biological disorder as is diabetes.

Stress by way of stress hormones
accelerate the shortening of Telomeres.

Genetic structures called Telomeres, protect the ends of our chromosomes from fraying. Our Telomeres shorten as we age thus we look old.

The length of Telomeres directly relates to the amount of stress one is under and amount of years they were under it.

Shortened Telomeres, decreased activity of this enzyme, and a very rough number for every year one was taking care of a chronically ill child, put on roughly six years of aging Co-discovered an enzyme (Telomerase) that can repair the damage to Telomeres. Preliminary data suggests that a meeting of minds, may have a health benefit by stimulating the healing effect of Telomerase.

Mothers of young children are a highly stressed group (work, child rearing, sometimes cant take care of themselves, add caring for a child with Sp. needs). It can tax the very reserves that sustain people, and they tend to die earlier.

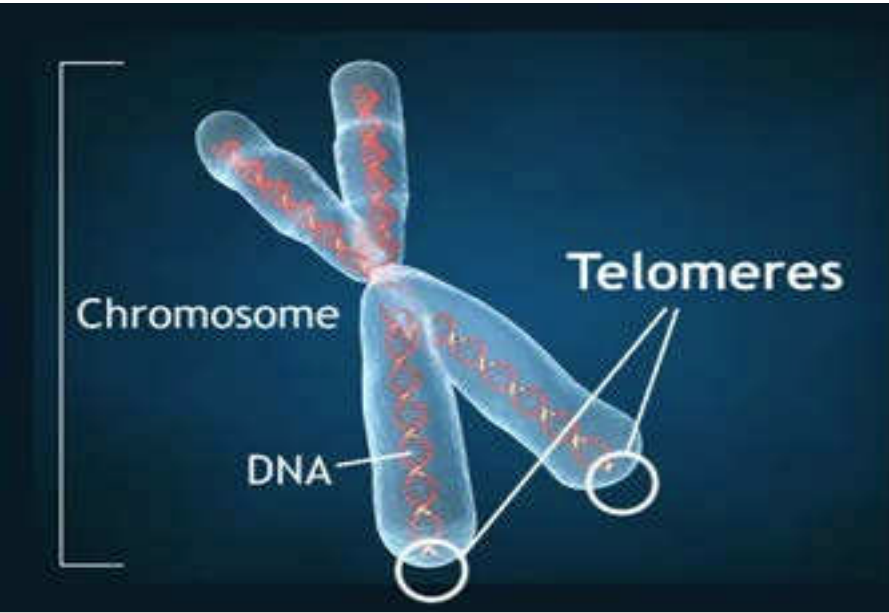
Compassion and caring for others, the meeting of minds, may be one of the most important ingredients that reduce stress and promote longevity, increase Telomerase, and keep cells rejuvenating and regenerating.

People belong to multiple hierarchies. One may have the worst job in the corporation, with no autonomy, control, and predictability, but being captain of the company's softball team that year (going to have all sorts of psychological means to decide: "its just a 9-5 job, it's not what the world is about, what it's about is softball, i'm the head of the team, people look up to me"). They are going to come out deciding they are on top of the hierarchy that matters to them.

The conditions in which people work and live are absolutely vital for their health. In general when people report that things have got worse, that their work stress has gone up, their illness rates go up.

Where people report that they've got more control, treated more fairly at work, more justice in amount of treatment, and things are getting better, the amount of illness goes down.

Giving people involvement in their work, giving them more say in what they're doing, and more reward for the amount of effort they put out; Creates not just a healthier workplace, but a more productive workplace.



The physical and social environment people work in is crucial for their health and productivity

Just 20-30 minutes of activity
most days of the week pays huge benefits
by lowering Cortisol every day and in the long-run.



Fear increases Cortisol.
Regular physical activity will
decrease fear by increasing
your self-confidence,
resilience, and fortitude—
which will reduce Cortisol.
Yoga will have similar
benefits with added benefits
of mindfulness training.

If your schedule is too hectic
to squeeze in a continuous
session of aerobic activity,
you can [reap the same
benefits by breaking daily
activity into smaller doses](#).
An easy way to guarantee
regular activity is to build in
adventant activity in to your
daily routine.

Try things such as riding
a bike to work, walking to
the store, taking the stairs
instead of the escalator...
These all add up to a
cumulative tally of reduced
Cortisol at the end of a day.

Mindfulness and Loving-
Kindness Meditation (LKM):
Any type of meditation will
reduce anxiety and lower
Cortisol levels. Simply taking
a few deep breaths engages
the Vagus nerve which
triggers a signal within your
nervous system to slow
the heart rate, lower blood
pressure and lower Cortisol.

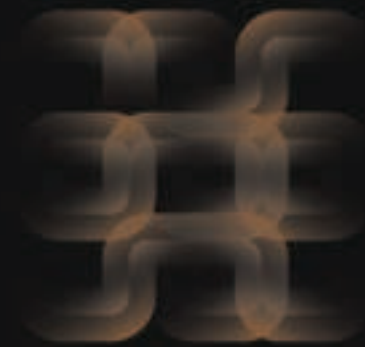
The next time you feel
yourself in a stressful
situation that activates your
'Fight-or-Flight' response
take 10 deep breaths and
feel your entire body relax
and decompress.



Keeping in mind to take 10 deep breaths can help your body adjust.



Taste



comfort for your senses

A study led by Sunil Kochhar of the Nestle Research Center in Switzerland, showed that people who rated themselves highly stressed to begin with had lower levels of stress hormones after eating 40g (1.4 ounces) of dark chocolate every day for two weeks (a little less than a regular-sized Hershey's bar).

More detailed findings in the Oct. 7 issue of the Journal of Proteome Research.

The study adds to a growing body of research showing that certain elements in chocolate — such as antioxidants called polyphenols — can have helpful health benefits.

Previous studies found chocolate can help fight heart disease and reduce the chances of developing cancer.

But moderation is key. Since most chocolate products contain fat and sugar, it's possible to have too much of a good thing.

Chocolates

Dark chocolate has been found to improve mood by increasing serotonin and endorphin levels in the brain.

According to research published in 2009 in the journal "Proteome Research" and reported by "The Huffington Post", eating 1.4 ounces of dark chocolate daily for two weeks is associated with lower levels of the stress hormone Cortisol. "Psychology Today" reports pregnant women who eat more chocolate during pregnancy give birth to more stress-free babies.

Darker chocolate has lower quantities of unhealthy ingredients, such as fat and sugar, than milk chocolate. [At least 70 percent cocoa Dark chocolate is the most beneficial variety, and it packs loads of nutritious antioxidants.](#) It even has numerous health-promoting properties that other kinds of chocolate do not contain; Dark chocolate has far more antioxidants and flavonoids than milk or white chocolate.

The flavonoids in dark chocolate —the antioxidant compounds that give it its bittersweet taste— promote a variety of cardiovascular benefits. The antioxidants in dark chocolate can also lower blood pressure, increase good cholesterol, decrease oxidation of bad cholesterol, which leads to plaque accumulation in the arteries, and promote blood sugar regulation. Dark chocolate can also lengthen the time it takes blood to clot and increase healthy blood flow by making blood vessels more elastic.

Do not replace any needed medication, for diabetes or hypertension, for instance, with dark chocolate.

A cup of hot cocoa has up to three times the antioxidants found in a cup of green tea and almost twice the amount in a glass of red wine. There are as many antioxidants in 1.5 ounces of dark chocolate as there are in five ounces of red wine.

Many fruits, vegetables, nuts, spices, chocolates, teas, wines, etc. Many of the fruits, nuts, vegetables and spices listed were analyzed for their Oxygen Radical Absorbance Capacity, known as ORAC. ORAC is one of several methods available to evaluate the antioxidant capacities of foods to keep harmful oxygen free radicals from damaging our bodies.

Milk chocolate, which most Americans eat, contains a low amount of flavonoids and antioxidant value compared to dark chocolate. It also has a relatively high sugar and saturated fat content and has been shown to increase blood cholesterol levels. White chocolate, which is a blend of cocoa butter and sugar, contains almost none of the antioxidants found in dark chocolate. In addition, more than half the fat in white chocolate is saturated.

Look for palm, coconut, hydrogenated and partially hydrogenated oils. These unhealthy oils should be limited or avoided, because they can raise blood cholesterol levels.



Antioxidants & Chocolates

Antioxidants can protect your body's cells from damaging oxidation much like oil or paint protects your car from rust.

Oxidation, which is the loss of an electron, sometimes produces free radicals that can cause oxidative stress or damage to the cells. Studies have shown antioxidants that neutralize this activity and that occur naturally in fruits, vegetables, green tea, garlic and onions can be effective at preventing DNA damage.

When free radicals damage the genetic material in a cell and it is not totally repaired, the damaged DNA is duplicated in new cells and can lead to these serious problems:

- Cancer
- Artery and heart disease (including heart attack, stroke, high blood pressure)
- Cataracts
- Age-related macular degeneration to eyes
- Diabetes
- Alzheimer's disease
- Gastrointestinal diseases
- Some degenerative changes that go with aging (e.g. brain function or cognitive impairment)



Free radicals that bombard your body cells come from high-fat diets; food additives; processed foods; soft drinks; infections; air pollution; cigarette smoke; ultraviolet light; radiation; excessive sun exposure; emotional stress; pesticides and many other environmental factors.

Tryptophan comes from serotonin, a neurotransmitter that influences mood. Low serotonin is both responsible for anxiety and caused by anxiety.

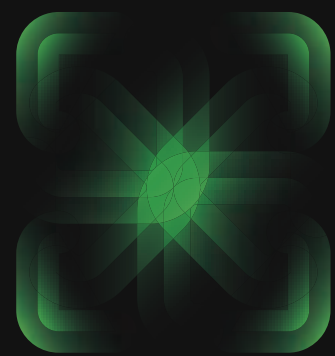
Theobromine is toxic to dogs, but studies have shown that it seems to have a positive, mood elevating effect on those that ingest it. Getting your magnesium through food is considered a much healthier overall option.

One thing worth trying is dark chocolate covered almonds. Almonds are incredibly healthy, rich in magnesium, and delicious. But don't simply depend on chocolate alone. Chocolate isn't going to magically cure you.





S m e l l



Peace for your senses

When groups of neurons are activated in the brain by any kind of stimulus — a picture, a sound, a smell, touch, taste, pain, pleasure, or emotion—a small electrical charge is generated, which indicates that neurological functions such as memory, attention, language processing, and emotion are taking place in the cortex.

Aromas can provide a soothing sensation to the entire central nervous system. If you find yourself caught up in a never ending cycle of work, stress, anxiety, and don’t know what to do, this may help you out.

It can take away pain throughout the body and some say it works by stimulating the lymphatic system. As with any form of treatment though, it’s all about preference and what works for you – so do some experimentation and see what you think works best.

These scents can be rubbed on to the wrists, temples, neck, behind ears, crown of head, upper back.

Avoid using these oils and products containing them before bed or they may make your insomnia worse.

Lavender & Jasmine



Within lavender there are over 150 different components that interact to help produce a healing effect.

Most people enjoy the scent of lavender and find that it really helps them de-stress and relax better than most. When inhaled it basically slows down fast wave brain activity (the root of anxiety) and replaces it with slower wave alpha brain wave activity.



Some have said Jasmine is just as good as the powerful anti-anxiety drug valium to help calm the nerves. The way that jasmine works is by increasing the GABA activity within the brain to help regulate feelings of stress, anxiety, and over-excitement.



Sandalwood & Grapefruit



Sandalwood has a sweet, woody, outdoors type scent and can be very beneficial in reducing anxiety. There was a study conducted that took into account the sandalwood aroma in cases of terminal illness. It was found to reduce overall anxiety in these patients. Additionally some have said that it works great to help with insomnia as well as improve overall quality of sleep. This fragrance comes from an evergreen tree



There are suggestions that grapefruit aromas can curb depression and enhance memory. Or taking a whiff before meals can control your appetite. Research from Japan's Osaka University shows that the scent of grapefruit actually boosts metabolism and reduces food cravings.



Sage Citrus & Amberwood



Sage is derived from Latin word "salvia" meaning "to heal". This oil should be avoided during pregnancy, by children younger than 5 years old. Additionally, those who have epilepsy or high blood pressure should not use this oil. Citrus oils also help to neutralise acid in your body and stimulate the immune system to fight infections. Another key feature of the citrus oils is that they have uplifting, invigorating and anti-depressant qualities. citrus essential oils contain one of the key components in these oils is an antioxidant known as limonene, and grapefruit has the highest percentage.

Blends with: Basil, bergamot, cedarwood, cypress, frankincense, geranium, juniper, lavender, peppermint, rosemary, rosewood and ylang ylang



The aroma of Amber has a pleasing effect that matches the ambience while practicing yoga and meditation. The aroma is long lasting and non-irritating. Amber oil when properly applied has a sedating effect and is very useful in inducing sleep. Studies have shown that Amber oil relaxes brain waves (alpha, beta and theta waves) through its psychoactive effect. It soothes the mind and helps fight problems such as depression, anxiety and stress. Europeans were among the first to recognize and credit amber's miraculous benefits for both mental and physical well being.

Blends well with: creamy vanilla and rich amber, with sandalwood.



Vanilla & Peppermint



The health benefits of Vanilla Essential Oil can be attributed to its properties as an antioxidant, aphrodisiac; causes excitement, anticarcinogenic; inhibits the development of cancer, febrifuge; reduces fever, antidepressant, sedative, tranquilizing and relaxing substance.

If you always thought that the word ‘Vanilla’ was the proprietary property of all those industries producing all those delicious pastries and sweet treats, well it’s time to reconsider.

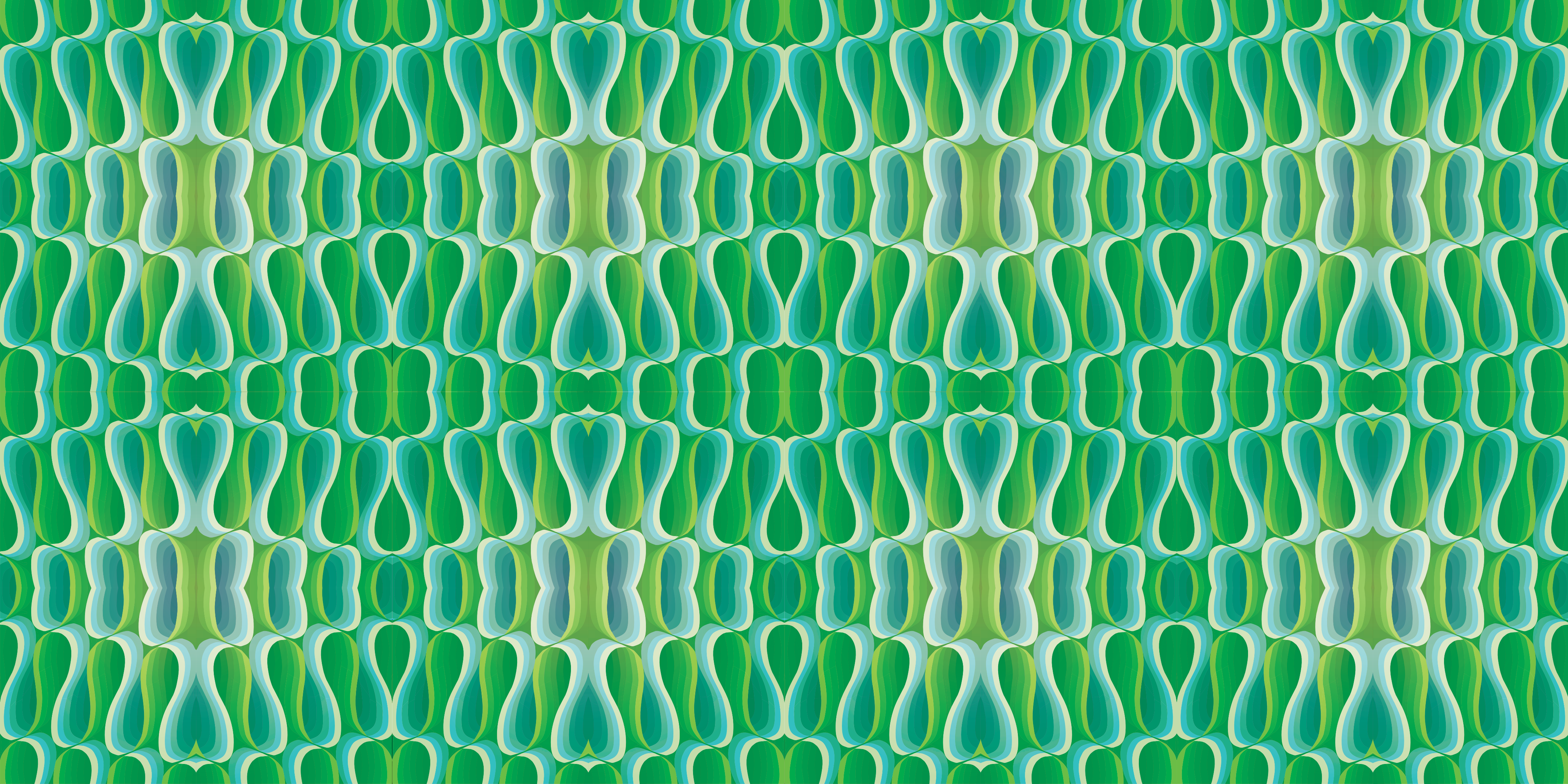


Peppermint oil is a natural stimulant that can increase your ability to concentrate during times of mental fatigue or stress. Inhaling the aroma can help alleviate jet lag and boost your energy levels. According to Janice Cox in her book "Natural Beauty at Home," peppermint oil can give you extra energy in the morning. It has mood-elevating properties that make it useful for alleviating symptoms of anxiety and depression.

It can help to boost your mood and relieve feelings of sadness as well as enhance mental performance. It also has the ability to improve circulation, heal the gut and relax tense muscles.

Avoid using peppermint oil as a stimulant daily, as it can interfere with sleep.





T o u c h



Delight for your senses

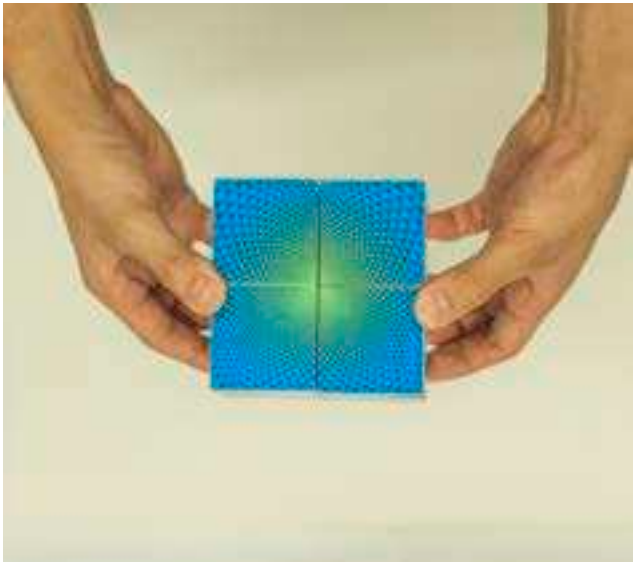
The cube that never ends include the visual aesthetics of calmness. There is a focus on curved shapes, lines, cool colors, and light, while fading was considered as secondary. Research on visual representations of the opposite of stress, which is de-stressing, calmness or tranquility; revealed that blue, water, and a sense of depth is present to visually remedy stress.

This not only is a visual experience to help counteract stressing, but a physical exercise in moving and stretching the whole hand and wrist. The cubes actually light-up when you touch the copper parts literally contributing to the calming aesthetic and a sense of control in the environment.

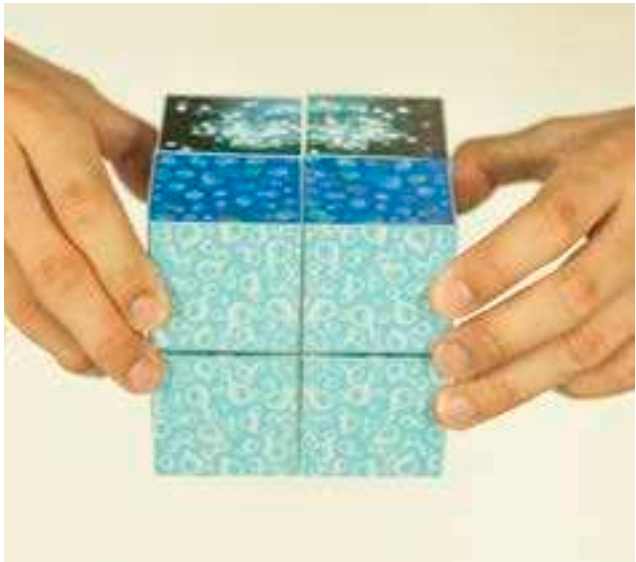
A short visualization is an easy way to get back to center. Even the National Institutes for Health recognizes the power of “guided imagery” to elicit a relaxation response. Instead of using words and music, this product uses visual representations to help spark imagination and creativity in referencing different things that contribute to relaxation and de-stressing.

Interacting with the cube

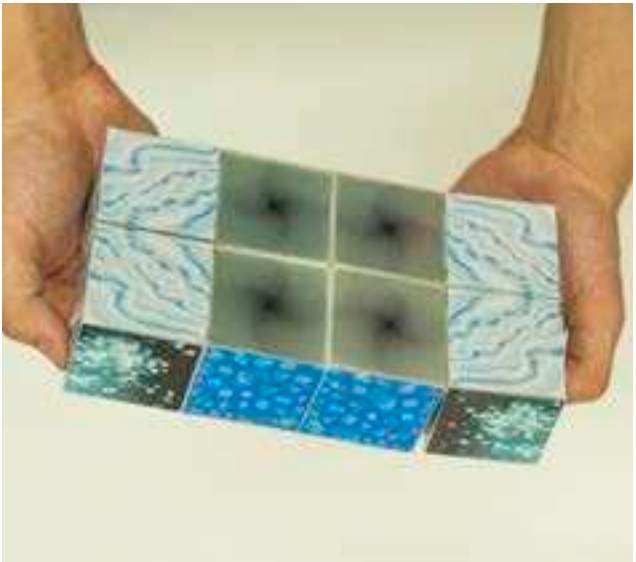
1/ The first pattern on top is an illusion of sorts, there appears to be a flower or circles that form.



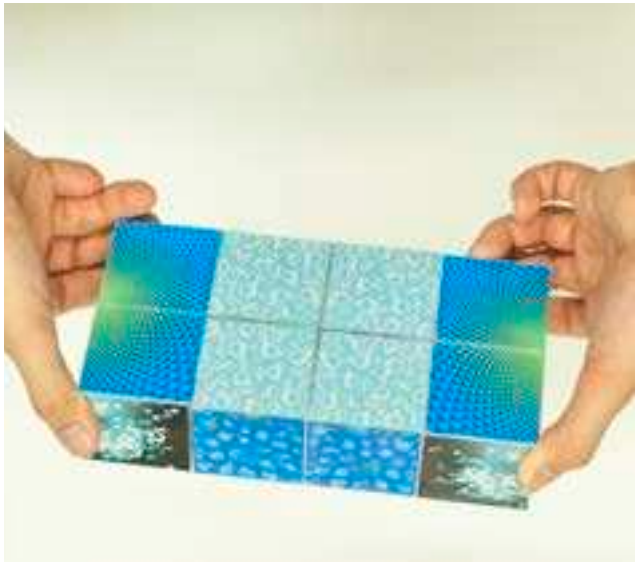
2/ To be sure to experience every combination of patterns, flip the cube over...



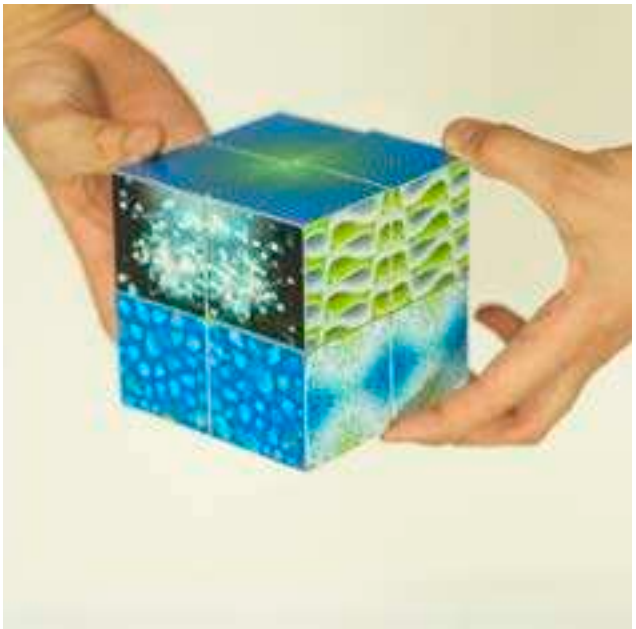
5/ The copper bits light up the cubes



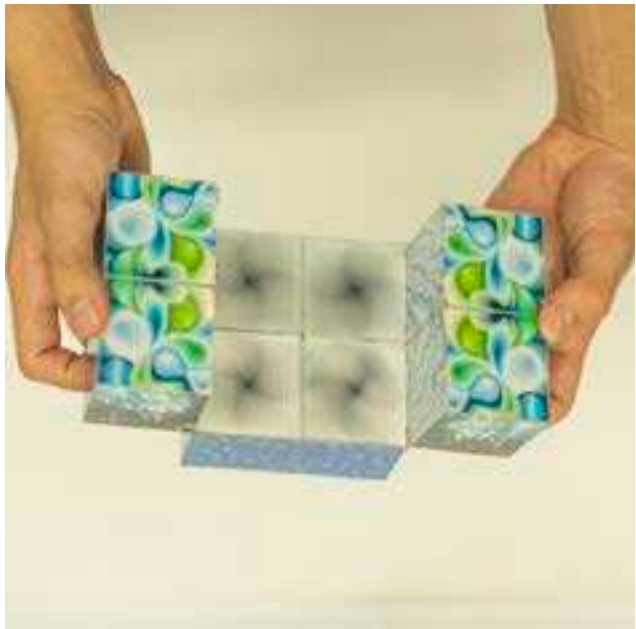
6/ Flip over the cube to see an alternative pattern combo on the other side.



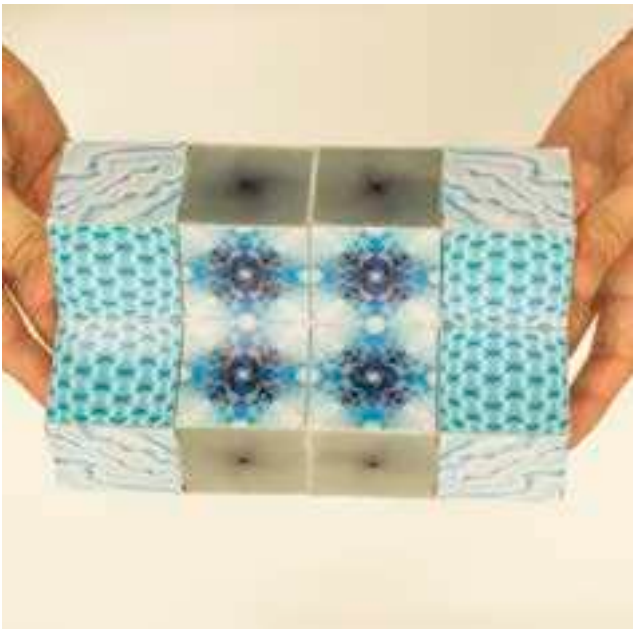
3/ ...and view the sides.



4/ Open the cube to the inside.



7/ Continue on to flip the cube and experience all sorts of patterns put together.

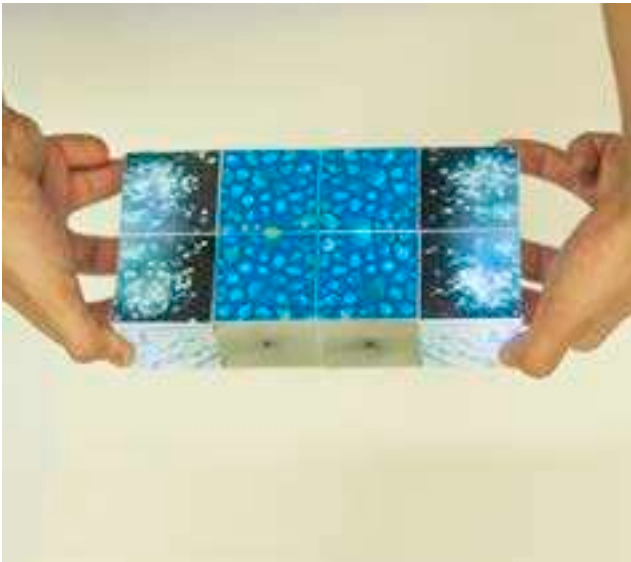


8/ It's best to see them in a dimly lit area.



Interacting with the cube

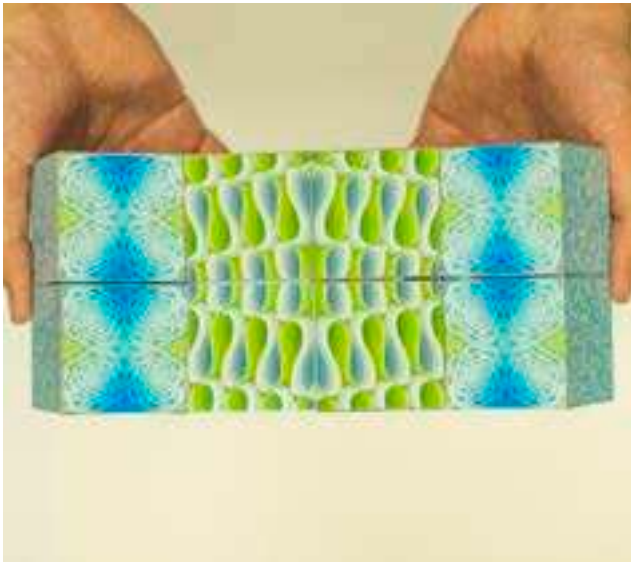
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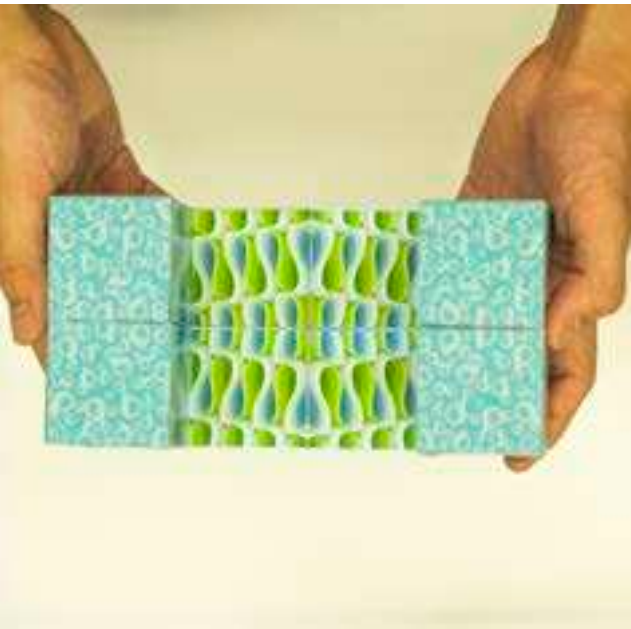
14/



11/



12/



Why are you encouraged to view this in a dimly lit room?

Being in a space with artificial lighting all day unnaturally elevates Cortisol at night, disrupting sleep and introduces a host of problems relating to body-fat levels, insulin resistance, and systemic inflammation. It also contributes to sleep debt and disrupts the neuroregulation of appetite. So at the end of the day try not to be in an extremely artificially bright environment.

Recently, scientists warned college students about the impact of light from computer monitors on Melatonin levels. They found that computer light at night — particularly blue light that was scanned by wearing different wavelength-blocking glasses — reduces college students' Melatonin levels. A related study found that backlit tablet displays suppress Melatonin.

Melatonin suppression is key to understanding much of why LAN (Artificial lighting at night) is so bad for us. This biochemical workhorse is produced by the brain's pineal gland at night. Regrettably, all this hormone and biochemical disruption is creating downstream effects — cancer being one of them.

Melatonin also has antioxidant properties, which plays an important role in anti-aging.

About the cube

Everyone knows that **broadly** moving and stretching slightly improves mood, even if for a moment. When stressed and rigid, our joints and muscles become tense and constrict blood flow, thus leading to soreness or pain if neglecting to remedy it for a while.

excessive muscular tone prevents optimal blood flow, changes local pH, and may cause shortening of the muscles over time.

But **fine** hand movements, repeated hour after hour, day after day, thousands upon thousands of times, eventually strain the muscles and tendons of the forearms, wrists, and fingers, causing microscopic tears. Injured muscles tend to contract, decreasing the range of motion necessary for stress free work.

The sheaths that cover delicate tendons run out of lubrication because they aren't given time to rest, so tendon and sheath chafe, resulting in pain. Due to this, tendons become inflamed, and begin to pinch neighboring nerves. This can result in numbness, tingling, or hypersensitivity to touch. Unless this cycle is interrupted, it repeats itself, and a long-term, chronic problem results.

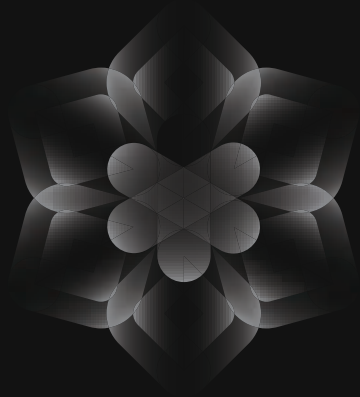
You may be at risk for developing an RSI (Repetitive Strain Injury) if you:

- Have poor posture
- Have poor technique
- Use a computer more than two to four hours a day
- Have a job that requires constant computer use, especially heavy input
- Don't take frequent breaks
- Are loose-jointed
- Don't exercise regularly
- Work in a high-pressure environment
- Have arthritis, diabetes, or another serious medical condition
- Keep your fingernails long
- Have an unhealthy, stressful, or sedentary lifestyle
- Weigh more than you should
- Don't sleep well
- Are afraid to ask for better accommodations
- Won't accept that you are at risk when you really are

The following checklist (adapted from Pascarelli and Quilter), can help you determine whether you have RSI. Do you experience:

- Fatigue or lack of endurance?
- Weakness in the hands or forearms?
- Tingling, numbness, or loss of sensation?
- Heaviness: Do your hands feel like dead weight?
- Clumsiness: Do you keep dropping things?
- Lack of strength in your hands? Is it harder to open jars? Cut vegetables?
- Lack of control or coordination?
- Chronically cold hands?
- Heightened awareness? Just being slightly more aware of a body part can be a clue that something is wrong.
- Hypersensitivity?
- Frequent self-massage (subconsciously)?
- Sympathy pains? Do your hands hurt when someone else talks about their hand pain?
- Avoid wearing or buying certain kinds of clothing because it is too difficult to put them on?
- Change shopping habits because you can't carry as much as you once could?
- Keep dropping things?

H e a r



Ease for your senses

Our brain’s ability to become flexible and/or transition through various brain wave frequencies plays a large role in how successful we are at managing stress, focusing on tasks, and getting a good night’s sleep. If one of the five types of brain waves is either overproduced and/or under produced in our brain, it can cause problems.

For this reason, it is important to understand that there is no single brain wave that is “better” or more “optimal” than the others. One particular brain wave will be dominant depending on the state of consciousness that you are in.

Brainwave entrainment is used by people who want to make a rapid change in their behaviour or state, and create lasting changes in their lives.

When you listen daily for a period of time, you are exercising your brain. After a period of use, you seem to be able to reach those areas of your brain at will to perform functions that you would not have been able to do before.

It is most beneficial to get into a routine and either listen when taking a break, at night before bed, or upon waking up in the morning.

Everyone can benefit so long as you are above age 17 and is not suffering from epilepsy or is wearing a heart pacemaker. You should also not be pregnant.

Do not listen while you are driving or working.

Brainwaves, Binaural Beats, Isochronic Tones

In traditional meditation practice, you are taught to focus on something to help still your mind and discipline yourself to remain focused. With brainwave entrainment audios and technology you don't need to maintain focus in order to reap the benefits because technology creates the same brain waves in mere minutes that traditional meditation may take years for you to achieve.

While using audio technology, electrical activity and energy patterns in your brain become more widespread throughout the brain instead of remaining confined to certain areas. When this type of stimulation to the brain occurs, your brain begins to create new neural pathways. The neural stimulation encourages new dendritic growth within the brain.

New dendritic growth allows for faster and smoother neural communication in the brain, and also provides more "processing power" in the brain. The dendrites are the many branching fibers extending from the neuron/cell body. These fibers increase the surface area available for receiving incoming information. The more dendrites the brain has at its disposal, the more quickly and smoothly it can process information.

Dendritic growth is a process that occurs throughout your lifetime. Whenever you learn something new, such as learning to play the piano, new dendritic growth occurs as a result of stimulating the mind in a new and different way. At the same time, old dendritic connections can become inactive and dissipate. (The shortening of dendritic branches and the reduction of the number of branches is associated with senility in the elderly.)

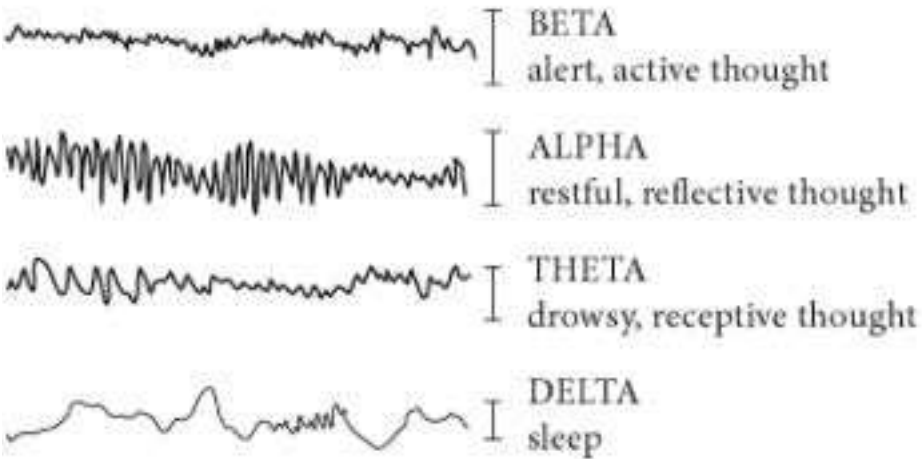
When you listen to binaural beats/tones, you are actually listening to two slightly different pitched frequencies. Each frequency goes to each of your ears independently. When this happens your brain responds by creating a third tone, making up the difference between the two frequencies. This shift in frequency then shifts your brain's consciousness and changes your state.

Isochronic tones are a powerful advancement in the field of brainwave entrainment. They are computer generated, rhythmically pulsed beats that safely, gently and effectively guide and entrain your brain to the most optimal brainwave frequencies.

You can choose the range of frequencies you want to experience. For example, if you want to focus attention to concentrate, you would choose an Isochronic tone on the alpha wave range (7-13 Hz). On the other hand, someone wanting to reach deep meditation would set them down into the Theta wave range (4-7 Hz).



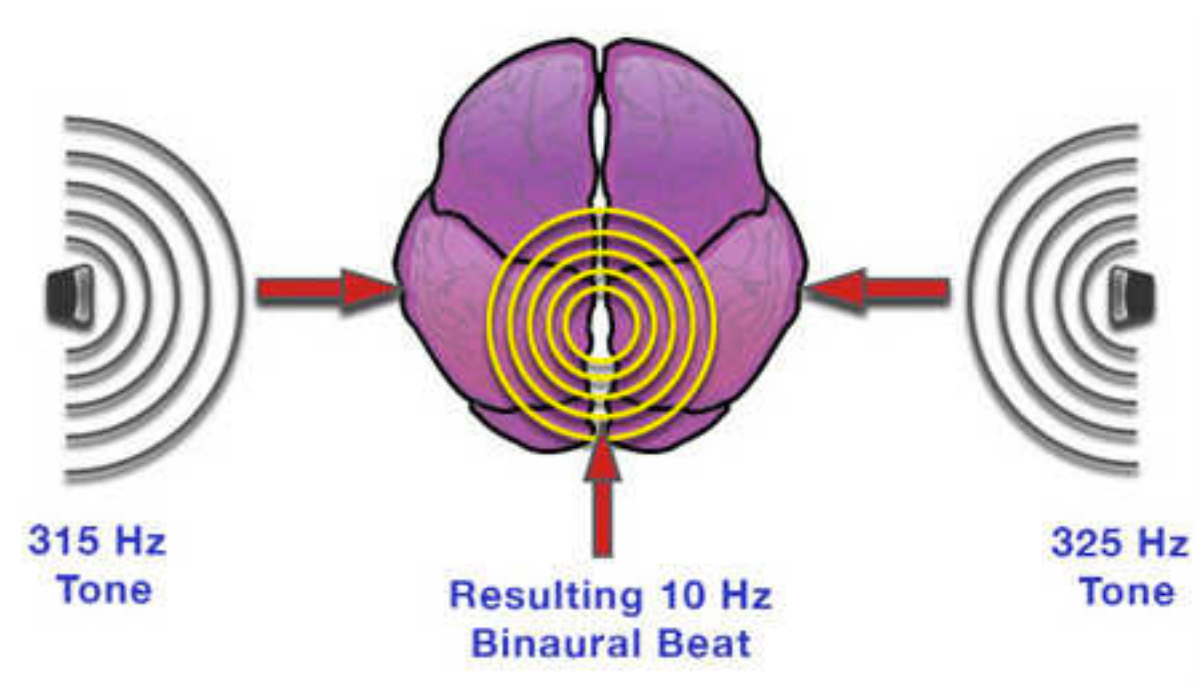
Growing Dendrites = Learning



Binaural Beats & Wave frequencies

How they work

Research has proven that introducing a binaural beat will cause the brain to begin resonating in tune with that beat. By creating a binaural beat at 10 Hz — (an Alpha frequency) you can trigger your brain to resonate at that same 10 Hz frequency, automatically inducing brain activity in the Alpha range. This same technique can be used to quickly and easily guide your mind into any state; called the Frequency Following Response.



Gamma Waves

These are involved in higher processing tasks as well as cognitive functioning. Gamma waves are important for learning, memory and information processing. It is thought that the 40 Hz gamma wave is important for the binding of our senses in regards to perception and are involved in learning new material; 100 Hz being the highest. It has been found that individuals who are mentally challenged and have learning disabilities tend to have lower gamma activity than average.

- Frequency range: 40 Hz to 100 Hz (Highest)
- **Too much:** Anxiety, high arousal, stress
 - **Too little:** ADHD, depression, learning disabilities
 - **Optimal:** Binding senses, cognition, information processing, learning, perception, REM sleep
 - **Increase Gamma waves:** Meditation



31 - 120 cps (cycles per second): Hyper brain activity, great for learning.

Wave frequencies

Beta Waves

These are known as high frequency low amplitude brain waves that are commonly observed while we are awake. They are involved in conscious thought, logical thinking, and tend to have a stimulating affect. Having the right amount of beta waves allows us to focus and complete school or work-based tasks easily. Having too much of this may lead to us experiencing excessive stress and/or anxiety.

The higher beta frequencies are associated with high levels of arousal. When you drink caffeine or have another stimulant, your beta activity will naturally increase. Think of these as being very fast brain waves that most people exhibit throughout the day in order to complete conscious tasks such as: critical thinking, writing, reading, and socialization.

- Frequency range: 12 Hz to 40 Hz (High)
- **Too much:** Adrenaline, anxiety, high arousal, inability to relax, stress
 - **Too little:** ADHD, daydreaming, depression, poor cognition
 - **Optimal:** Conscious focus, memory, problem solving
 - **Increase beta waves:** Coffee, energy drinks, various stimulants

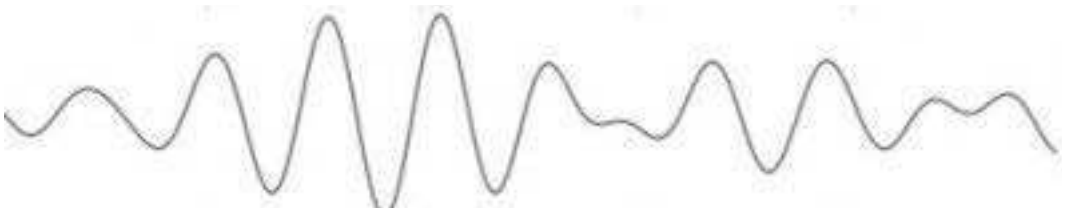


13 - 30 cps (cycles per second): Busy engaged in activities and conversation.

Alpha Waves

This frequency range bridges the gap between our conscious thinking and subconscious mind. In other words, alpha is the frequency range between beta and theta. It helps us calm down when necessary and promotes feelings of deep relaxation. If we become stressed, a phenomenon called “alpha blocking” may occur which involves excessive beta activity and very little alpha. Essentially the beta waves “block” out the production of alpha because we become too aroused.

- Frequency range: 8 Hz to 12 Hz (Moderate)
- **Too much:** Daydreaming, inability to focus, too relaxed
 - **Too little:** Anxiety, high stress, insomnia, OCD
 - **Optimal:** Relaxation
 - **Increase Alpha waves:** Alcohol, marijuana, relaxants, some antidepressants



8 -12 cps (cycles per second): Very relaxed. Deepening into meditation.

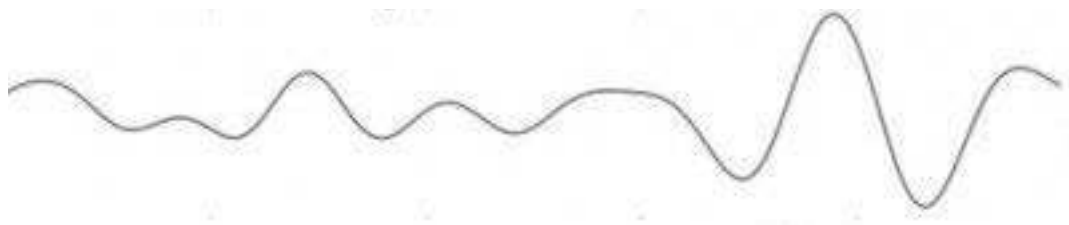
Wave frequencies

Theta Waves

This particular frequency range is involved in day-dreaming and sleep. Theta waves are connected to us experiencing and feeling deep and raw emotions. Too much theta activity may make people prone to bouts of depression and may make them “highly suggestible” based on the fact that they are in a deeply relaxed, semi-hypnotic state. Theta has its benefits of helping improve our intuition, creativity, and makes us feel more natural. It is also involved in restorative sleep.

As long as theta isn’t produced in excess during our waking hours, it is a very helpful brain wave range. Research has proven thirty minutes a day of Theta meditation can dramatically improve a person’s overall health and well-being. Theta meditation has also been known to cause the need for less sleep.

- Frequency range: 4 Hz to 8 Hz (Slow)
- **Too much:** ADHD, depression, hyperactivity, impulsivity, inattentiveness
 - **Too little:** Anxiety, poor emotional awareness, stress
 - **Optimal:** Creativity, emotional connection, intuition, relaxation
 - **Increase Theta waves:** Depressants

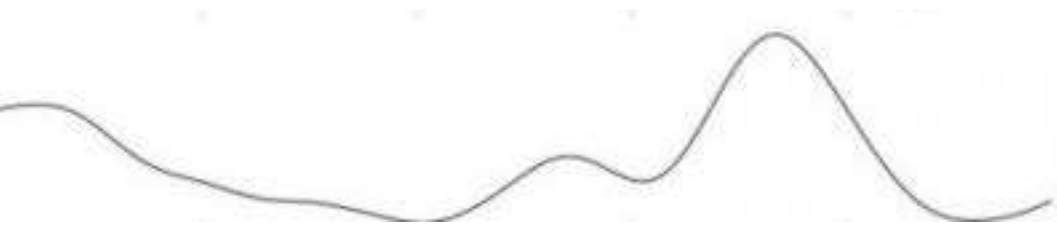


4 - 7 cps (cycles per second): Drowsy and drifting down in to sleep and dreams.

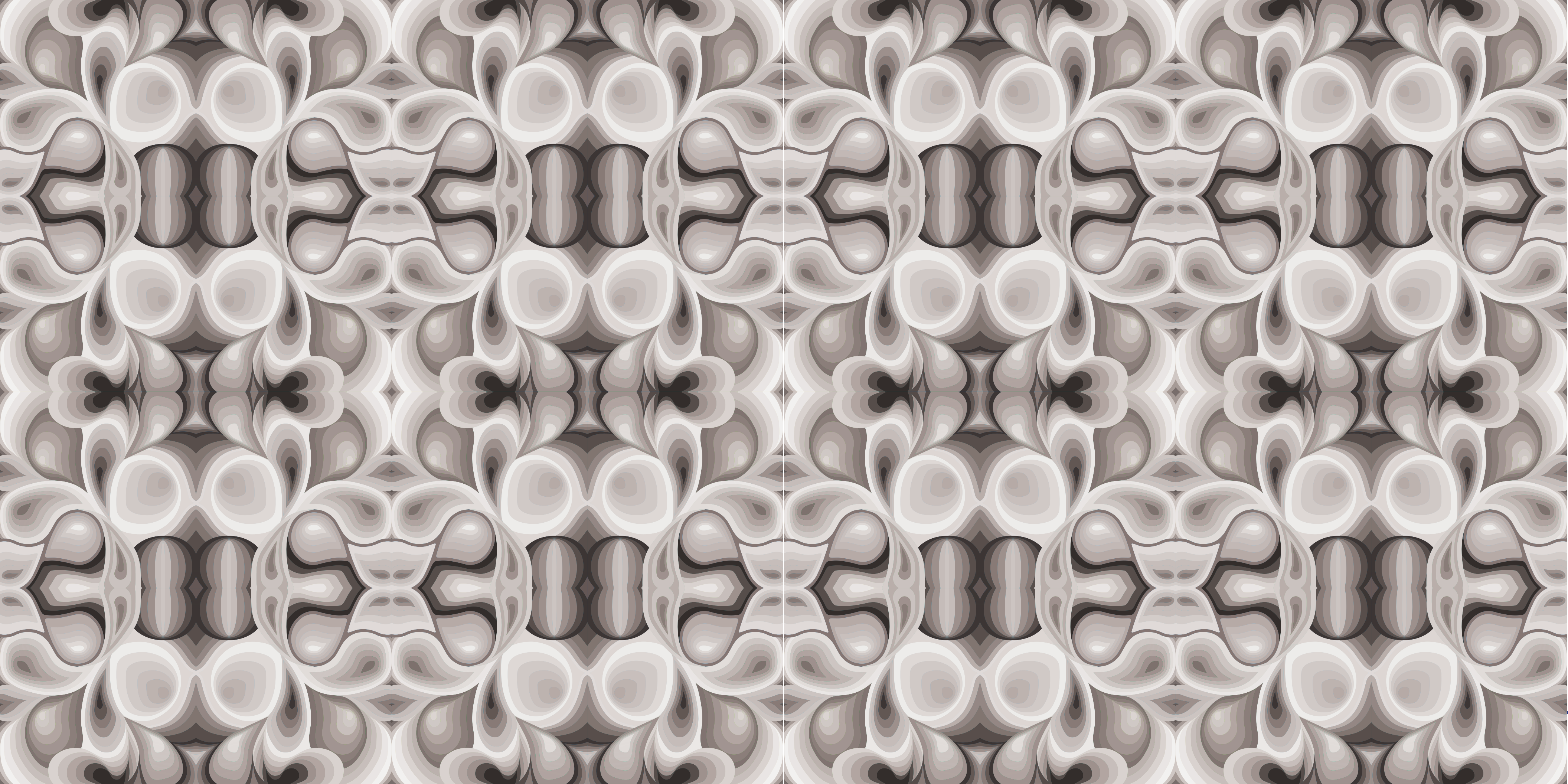
Delta Waves

As we age, we tend to produce less delta even during deep sleep. They are associated with the deepest levels of relaxation and restorative, healing sleep. They have also been found to be involved in unconscious bodily functions such as regulating heart beat and digestion. Adequate production of delta waves helps us feel completely rejuvenated after we wake up from a good night’s sleep. If there is abnormal delta activity, an individual may experience learning disabilities or have difficulties maintaining conscious awareness (such as in cases of brain injuries).

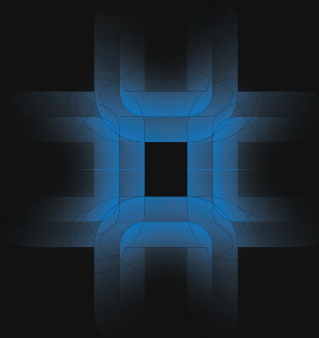
- Frequency range: 0 Hz to 4 Hz (Slowest)
- **Too much:** Brain injuries, learning problems, inability to think, severe ADHD
 - **Too little:** Inability to rejuvenate body, inability to revitalize the brain, poor sleep
 - **Optimal:** Immune system, natural healing, restorative /deep sleep
 - **Increase Delta waves:** Depressants, sleep



5 -3 cps (cycles per second): Deeply asleep and not dreaming.



View



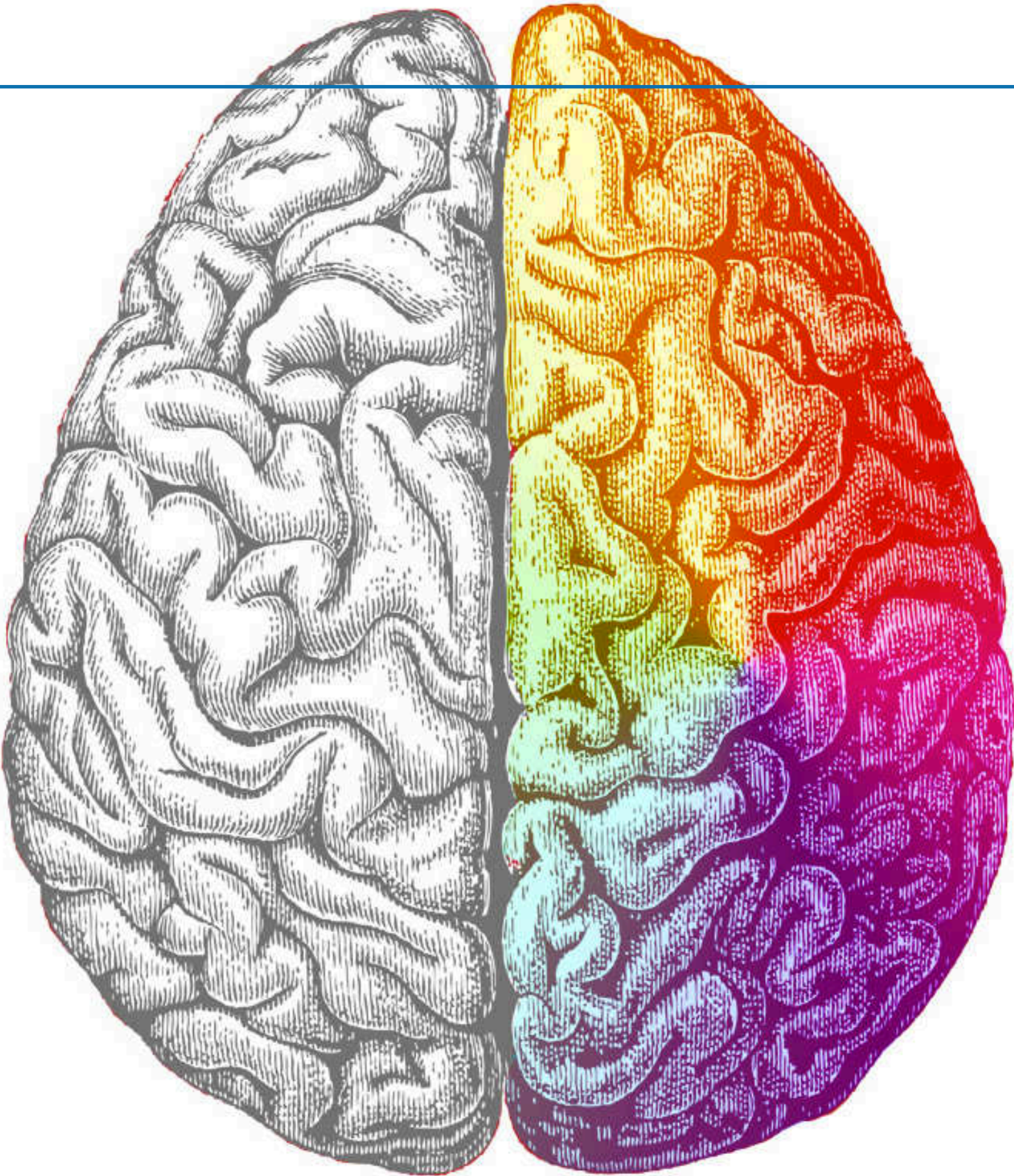
Beauty for your senses

Working a puzzle helps develop your abilities to reason, analyze, sequence, deduce, logical thought processes and problem solving skills. These types of puzzles also improve hand-eye co-ordination and develop a good working sense of spatial arrangements.

When you solve a jigsaw puzzle, you are engaging your brain to retain information on shapes and colors in order to choose pieces that will fit together properly. This hunt for pieces requires your brain to memorize what each piece looks like or what the shape should look like and what kinds of pieces you are searching for in order to complete the picture. Because you are likely doing this repeatedly as you search for puzzle pieces, the process reinforces short-term memory.

Jigsaw puzzles also promote the relationship between the left and right brain. The logical left brain looks at individual parts. It is sequential, rational, analytical and objective. The left brain is stimulated by problem solving. The creative right brain sees the big picture. It thrives on randomness, is intuitive and subjective, and even likes the unfamiliar. Jigsaw puzzles satiate the needs of both the left and right brain.

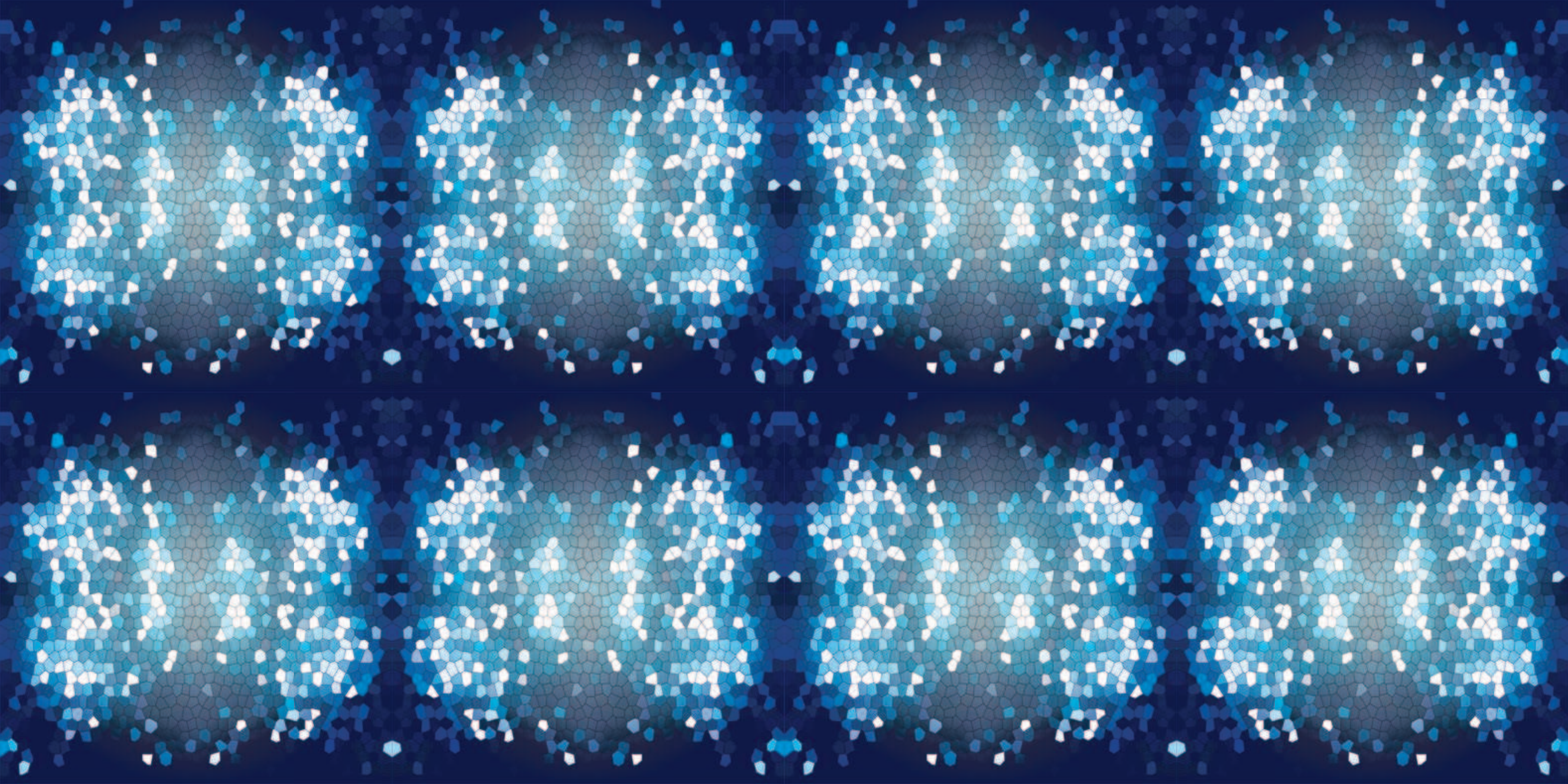
In solving jigsaw puzzles, the brain is being worked in both hemispheres, making connections between the sides as well as between brain cells. The connections enhance your ability to learn, understand and remember.



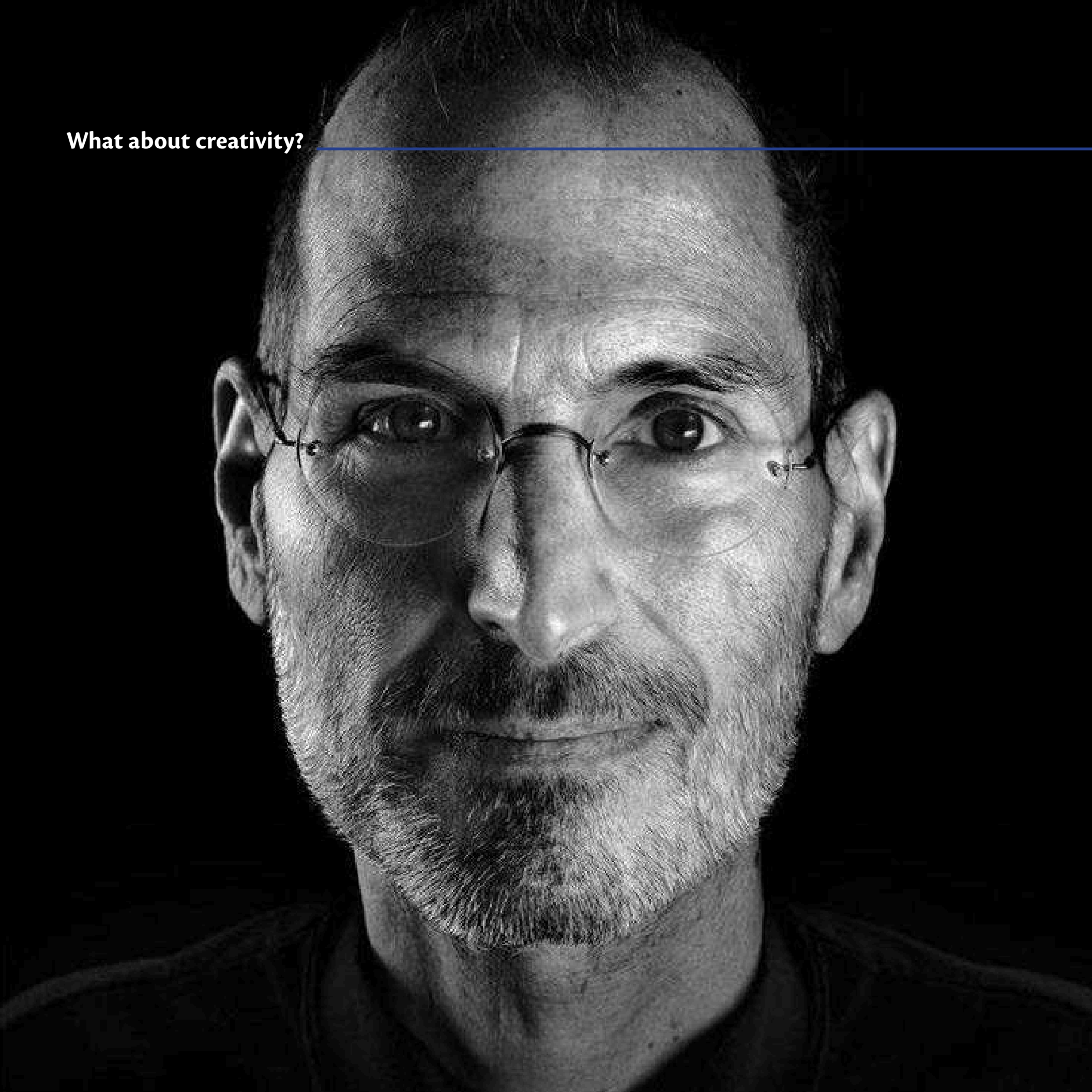
Each success with the puzzle – from individual piece placement to the actual completion of the puzzle – encourages the production of dopamine, an important neurotransmitter in the brain that regulates mood and affects concentration and motivation. Dopamine plays a huge role in memory and motor control.

Puzzles can also be relaxing. Concentrating on the puzzle pieces might help refocus the mind away from negative and stressful thoughts. Once you're in a peaceful state of mind, Cortisol and blood pressure levels drop to healthier levels. Since chronic elevations of Cortisol in the bloodstream can lead to impaired cognitive abilities, anything you can do to reduce those levels can only be beneficial for your brain and overall health.

Working with Jigsaw puzzles helps our physical health by lowering our breath rate reducing heart rate and blood pressure. keeping the mind active with Jigsaw puzzles and other mind-flexing activities can provide a longer life expectancy, a better quality of life.



What about creativity?



“Creativity is just connecting things.

When we ask creative people how they did something, they feel a little guilty because they did not really do it, they just saw something. It seemed obvious to them after a while. That’s because they were able to connect experiences they’ve had and synthesize new things. The reason they were able to do that was that they’ve had more experiences.”

Steve Jobs

What about creativity?

The deliberate and spontaneous pathways to creativity

The main difference is the executive center (especially in the left hemisphere), remains steadfastly in control of the creative process in the deliberate pathway. The Executive Center directs what you think about and what is called up from your memory bank.

The spontaneous process allows more integration from the non-dominant hemisphere of the brain (right hemisphere for right-handed people, even many left-handers). Resulting in a creative idea suddenly springing into consciousness when least expected.

The benefit of the spontaneous pathway is the possibility of more novel and more abundant ideas. In spontaneous thoughts, multiple ideas using a wider database can be processed at the same time.

When thinking consciously and deliberately, your attention is focused and you can only process one thought at a time.

The seven brainsets to creativity.

Absorb: Receptive to more info coming in from senses from the environment and to information from within your own thought processes (usually unconscious)

Envision: Imagination. Think visually and with your senses rather than verbally. Seeing and manipulating objects in your mind. Use the mental format of “What if”

Connect: Divergent thinking. Generating multiple uncensored solutions to open-ended problems. Allowing you to see connections between objects or concepts that are quite far from each other.

Reason: Constantly manipulate info in your memory to solve problems in a logical and sequential way. Trial-and-error development

Evaluate: Constantly judge the value of ideas, concepts, products, behaviours, or individuals. Necessary for deciding which of your creative ideas are worth pursuing and constantly monitoring their usefulness and appropriateness

Transform: A self-conscious and dissatisfied/distressed state of mind. Although vulnerable, you are motivated to express in creative form the anxieties, and the hopes we all share as part of the human experience.

Stream: State of flow. Thoughts and actions begin to stream in steady harmonious sequence, almost as if orchestrated by outside forces. Improvisation to make creative material.

What about creativity?

Absorb Brainset

If not willing to access this state, its difficult to be innovative and creative. Instead one would be using the deliberate processes (often effective, tiring, time-consuming).

Associated with words such as autohypnosis, trance, alpha state, absorption, mindfulness, primary process thinking, openness to experience, dissociation, and transliminality.

Ellen Langers cognitive theory of “mindfulness” associated with openness (flexible frame of mind). Mindlessness is characterized by pigeonholing experiences, ideas, and people into rigid categories.

Premature cognitive commitment –identifying/ labeling a chair as where one sits –reduced the probability of seeing its myriad other realities.

When entering the absorb brainset, beta activity is reduced, and brainwaves exhibit slower frequency (5-12 Hz) and higher amplitude called alpha and theta activity. (Alpha & Theta activity are associated with a relaxed and receptive mental state)

Most important strategy for thinking and acting creatively is to develop your ability to absorb information nonjudgmentally.

Results from the absorb brainset:

- You notice new things.
- More receptive to seeing associations between things in the environment and problems you’re trying to solve.
- Remain receptive to ideas originating in your unconsciousness.
- Can’t make novel and original combinations out of information which you aren’t aware.

When in the absorb brainset, be aware of your environment and try to notice shapes, curves, objects; why are they the way they are? is there any aesthetic value? How do you feel when looking at your environment? Absorb your environment and create snap-shots or notes.



What about creativity?

Envision Brainset

The last part of the brain to develop is the prefrontal cortex (as you mature). (Also not entirely developed until around age 25).

Facilitates imagination, provides link between deliberate and spontaneous pathways to creativity by allowing you to retain purposeful and deliberate control of the contents of conscious awareness while accessing information to ideas that are developed at more spontaneous and disinhibited levels of cognitive processing.

In order to use mental imagery in a creative manner its important to practice both seeing and manipulating objects.

Hypothetical thinking involving “What If”.

Einstein claims to have used “what ifs” to form his theory of relativity: his creative process seeing more or less clear images which can be voluntarily reproduced and combined (essential feature in productive thought).



Imagine and manipulate your environment while saying “what if”

Traditional methods (some are not recommended) of regulation include: dreaming, sleep deprivation, alcohol, meditation, repetitive or rhythmic exercise (long-distance running, trance dancing).

Main pattern of activation for the envision brainset is a network connecting the executive centers (particularly in the right hemisphere) to left association centers and to areas in the parietal, temporal, and occipital lobes dedicated to processing information from your senses.



What about creativity?

Connect Brainset

Think divergently rather than convergently. Make connections between disparate objects and concepts: leading to more novel and original ideas.

Scientific evidence that indicates you're more likely to generate a large number of ideas and to make unusual associations when you receive an unexpected reward or when you're in a good mood.

Another test found that people score better on divergent thinking tasks after true laughter or a good joke.

The last part of the brain to develop is the prefrontal cortex as you mature. (Also not entirely developed until around age 25).



Try and find humor in your environment, or be around positive people.

Rapid thinking associated with divergent thinking can increase positive mood and creativity. Motivation to work on and complete the project is enhanced.

Positive mood and the self-esteem that accompany lead to an increase in agency, or the belief that you have the power to overcome obstacles and achieve your goals.

Positive mood allows for more playful and divergent thinking because it signifies that all is well –your mind is not occupied with a survival mindset.

Connect Brainset Cont.

Spending more time outdoors in the sunlight might have the same effect on divergent thinking.

Spending time in beautiful scenery releases endogenous opioids that increase positive mood and decrease cognitive inhibition. (easier utilizing the connect brainset)

Being in a good mood does not evoke divergent thinking and the connect brainset. You must change your mood into a positive direction to get this effect. (listening to upbeat music, hearing a good joke, engaging in physical activity).

There is a lack of research on the effects of caffeine on creativity. One study found artists tend to use more caffeine than other members of a control group.

Small doses of caffeine have a beneficial effect, while higher doses hamper effective thinking.

Activation of the right hemisphere is known to evoke broad attentional focus, whereas left hemisphere activation evokes a narrowing and focusing of attention.

The connect brainset is characterized by defocused attention, increased right hemisphere activation, and increased activity in the association areas of the brain.

Beneficial when coming up with more than one option for creative problem finding or problem solving. Most beneficial when trying to find solutions to ill-structured open-ended creative problems.

What about creativity?

Reason Brainset

The activation pattern for convergent thinking.

Although your brain is not limited to thinking convergently here; it can also tackle ill-structured or open-ended problems. Does so by generating one possible solution at a time and using trial-and-error techniques, rather than the generation of multiple solutions such as in divergent thinking.

Tend to think in realistic and practical terms. The Executive center makes plans sequentially. Planning for the future is basically just setting goals. Goals are essential to creative work.

Goal setting can help you achieve your creative objectives: Motivate action, help you manage time, increase chances for success, increase self-confidence, increase sense of control over your life.

The reason brainset is an active brainset (requiring a certain level of physiological and mental arousal). An increase in the so-called arousal neurotransmitters: including glutamine, dopamine, and norepinephrine. A number of methods to effect this arousal: most effective and important is moderate physical exercise. Taking a walk will reliably activate the frontal lobes of your brain. As well as Coffee and music (classical).



What about creativity?

Evaluate Brainset

A brain state which you are focused on judging an aspect of your environment.

Decreases flexibility and closes doors. Decreases the possibility that a thing could be other than which is dictated by the pigeonhole into which you have put it.

Active Judgment:

The goal of making judgments is to improve our ability to predict the future.

A human tendency to categorize and judge things and experiences. (putting them in pigeonholes so that we know what to expect from them if we encounter them again).

Categorization is a form of judgment that improves our future speed of decision making based on past experience.

Know what constitutes a good idea and a bad one: One way to do this is to learn as much as you can about the domain or area of creativity in which you're working.

Know the "rules" of the domain and how experts in that area evaluate creative work: takes time and experience in that domain which a 10-year rule applies.

Remain impersonal. You are evaluating an idea or piece of work, not yourself or your abilities

Give yourself freedom to fail

If you start feeling low or discouraged about your idea or product, listen to the running commentary of thoughts in your mind-psy-
chologists call this "self-talk"

Is there any evidence for your negative thoughts? Gather evidence for and against your negative self-talk with more positive and realistic statements:

"well that idea that won't work, on to the next one"

"that was a mistake - but if I don't make mistakes I won't learn anything"

If ideas pop into your head, you can measure them against the standards for all creative products:

Is the idea original?

Is it expressive in a way that will be useful to someone?

Will this idea change the way people think about things?

Does the idea meet the goal that I set ahead of time?

If not, does it meet a goal that might be useful to pursue at another time?

Is it beautiful (does it have aesthetic value)?

What about creativity?

Transform Brainset

State of consciousness where your attention is occupied by thoughts relating to yourself and your feelings, which are generally negative. Emotions color the way you see your environment, way you recall memories, and all aspects of your cognition. Can get in the way of creative efforts, or can enhance your creativity.

A Low level background feeling state (not consciously aware of feelings) is referred to as “Stream of Affect” [by psychologists David Watson and Lee Anna Clark] “Affect” being a psychological term for emotions.

Positive stream of affect makes us somewhat more open to novelty, negative stream of affect somewhat closes us up to new ideas.

Next step up in emotional intensity is mood. (feeling anxious, irritable, or “blue”). This occupies more of your conscious attention. Moods such as anxiety or depression can make it difficult to concentrate.

The most intense emotional state you can experience is the onset of an actual emotion. Lasts relatively a short duration, less than an hour, and is a response to a particular trigger (in the environment or an internal event), but is extremely intense.

Accompanied by characteristic facial expressions, thoughts, physiological changes, and subjective feelings. Also prime the body to behave in specific ways called “action tendencies” (ex. Fear – fight or flight, anger – aggression).

If the emotion is intense enough, you may feel completely controlled by it. This is called “emotional hijacking”.

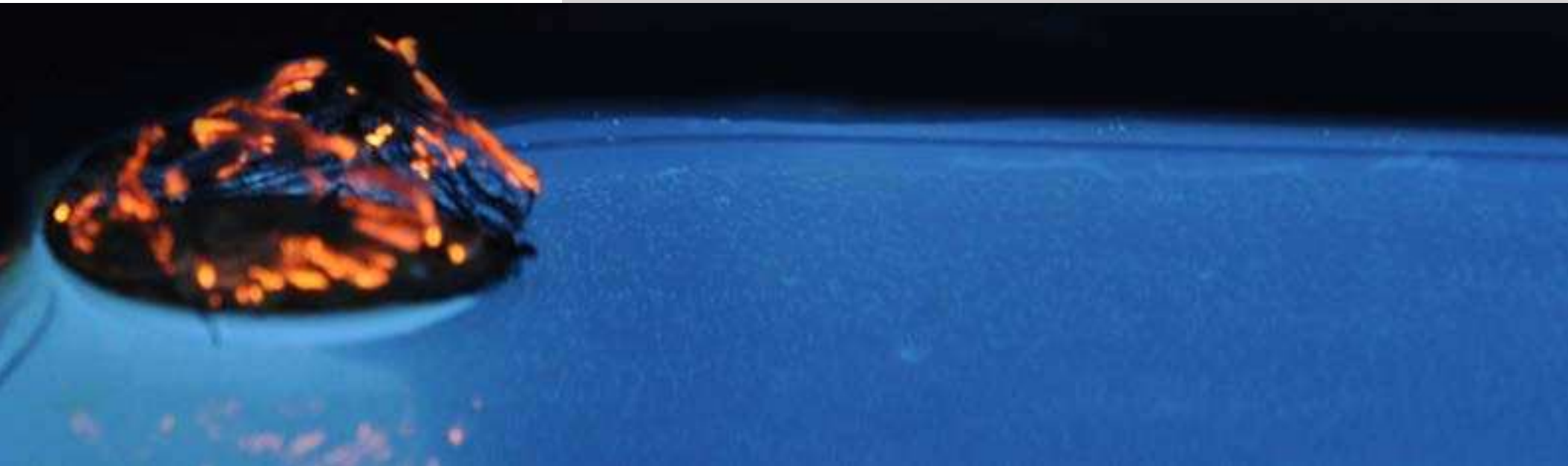
When emotionally hijacked your action tendency becomes an action imperative.

Fear and depression are a turnoff for the idea-generating brainsets discussed. (If you are in a negative feeling state, you’re in the transform brainset)

Highly creative individuals are neurotic (experienced long bouts of anxiety, irritability, or depression). They have found ways to use their state of negative affect to enhance creativity:

1/ using creative work to assuage their negative feelings; the negative feelings act as a motivator to be creative and creative work acts as a self-administered therapy

2/ using negative feelings as subject matter for their creative work.



In the creative process, the negative stream of effect is best left with its embers.

What about creativity?

Transform Brainset continued

Your internal world is where all creative ideas are generated and also where all psychic healing takes place. While you're there, how about doing a little exploring and investigating? –It can yield an enormous amount of knowledge about yourself and about the human condition in general.

Self investigation should take place in a state of mild dysphoria or negative mood. If feeling hopeless or out of control, you should not perform the transform exercises.

It is Important to recognize you have enormous strengths as well as flaws. To focus on the negative aspects of your character is to disregard half your essence.

If we don't learn that we can endure negative feelings- and come out the other side stronger- we will be subject to episodes of depression whenever the normal vicissitudes of life deal us a blow.

In psychoanalytic terms, the energy of negative emotions that is too dangerous or unacceptable to be expressed directly can be redirected into creative work that is more socially acceptable (defense mechanism called sublimation).

As an individual releases negative energy into creative work, the power of negative emotions is weakened.

Misery *loves* company.

What about creativity?

Stream Brainset

The brain activation state for what psychologist Mihaly Csikszentmihalyi calls “flow”.

This is where one loses both sense of self and sense of time as they build a set of spontaneously and skilfully executed responses to a challenge.

The sum of the responses results in an improvisational performance that researchers call “ecologically relevant creative behaviour”; the creative behaviour has been shaped in part by your relationship to your current environment.

Demonstrated by jazz musicians improvise actors, neurosurgeons, novelists who write automatically.



This occurs when the challenge of the activity requires all your attention and there are no attentional resources remaining with which to process other stimuli. Distractions go unnoticed. All attention is directed to the task, and there is no worry of failure. Attention is focused on the here and now, and there is no room for concern.

Self-consciousness disappears. The boundaries of the ego seem to dissolve and you become part of something bigger than yourself, in harmony with the universe.

Time becomes distorted. Time may speed up, or time may slow down. The activity becomes an end in itself. The activity becomes intrinsically rewarding than being a means to an end.

What about creativity?

Stream Brainset continued

The expertise has to be appropriate for the type of challenging work which engaged in. But the components of that expertise have to be represented in the memory networks of the brain.

The Implicit memory contains info that you've learned to the point it has become automatic. Also referred to "overlearning" or "muscle memory". Generally one can not describe the contents of implicit memory with words, but can access it without consciously thinking about it.

Explicit memory includes info that you can access consciously. Most of what you learned at school is represented in explicit memory, as are your personal autobiographical memories.



To increase intrinsic motivation in the activities filling your daily life:

- 1/ Spend more time doing activities you truly enjoy.
- 2/ Increase motivational salience of the activities you are already doing.

If the task seems too challenging for your skill level:

- 1/ Break the task down into smaller manageable parts.
- 2/ Reinforce your skills. (Postpone a significant challenge if you feel that you don't have the implicit resources to handle it. However, it's more likely you just need to remind yourself of your own existing resources.)

If the task seems too simple, there are several ways to kick the challenge up a notch:

- 1/ Set time limits on the task or establish deadlines.
- 2/ Increase your appreciation for the task. (step by step analysis and appreciation/gratitude for each step or item)
- 3/ Increase your standards for the task performance.

What about creativity?

Ending notes

For most people, regardless if they're left or right handed, the left hemisphere is the dominant hemisphere. Specialized for language, analysis, details, and sequential thinking.

Right hemisphere is specialized for spatial skills, analogies, big picture, and nonsequential thinking. In normal convergent thinking, the left hemisphere is more highly activated than the right, often to the extent that right hemisphere ideas and skills don't get represented in conscious awareness.

High levels of the neurochemicals Norepinephrine and Dopamine allow to focus in on details, whereas lower levels may be associated with encoding the broad overview of an event. (Reduced memory for details may be due to partial deactivation in these brainsets of the left executive center, which is specialized in processing details. Also may be due to fluctuations in norepinephrine).



The flexibility to shift brainsets is associated with higher overall cognitive functioning. A group that is able to shift back and forth between the connect and reason brainset (divergent and convergent thinking) get better over time about thinking divergently, and the group not able to shift repeatedly gets worse at divergent thinking over time. Both groups maintain their original thinking level.

Doing activities that guide you into different brainsets automatically affects the flow of neurochemicals. Another way of affecting your brainset is through learning. Increasing your knowledge base and learning new skills, you will also improve brain state flexibility and ward off cognitive decline in your later years.

It is important to keep on using and challenging your brain, or at least learning new things. Just as the tree continually takes in nutrifice from it's environment and grows, your brain will grow (Dendrites – pg. 56) just like it.

What about creativity?

Final ending notes

Continuing to learn has the benefit of reversing cognitive decline and slowing the progression of dementias such as Alzheimer's.

Two types of learning that are going to help you become more creative:

1/ learn about many different types of things; increasing general knowledge; keep folders; should become a way of life.

2/ learn a new skill.
Stimulates different parts of the brain than does semantic knowledge.
(if you alternate focusing on a new skill and practicing skills you already possess, you will be also getting practice in brain state flexibility).

(the worlds most creative people have interests in multiple domains:
poly=many + mathes = to learn; polymaths)

Continue on travelling to new places, learning, and re-learning, and then you will continually be flexible and progress more.



“A lot of people in our industry haven't had very diverse experiences...

So they don't have enough dots to connect...

and they end up with very linear solutions without a broad perspective on the problem.

The *broader* one's understanding of the human experience, the better design we will have.”

- Steve Jobs



What students say
about stress

“A *short* walk with
my dog is my
favourite way to take
a break and destress!

...And it's the most
effective.”

“I find near the tail
end of my stress
period or anxiety
feelings that my
creativity gets a boost

...and i’m able to see
situations or new things in a
different, often more open
minded light than before.”

“I get stressed
by people who
don’t take time
to understand art

...style, progression,
and other people.”

“Stressful situations
mentally pushes
me to think more
creatively

...but I take less care
of myself physically.”

“I used to thrive in
stressful situations
in high school

...but now I completely
crumble under pressure.”

“Collaborative work
provides interesting
potentials because
we share the stress

...and the project becomes less
of a life and death problem.”

“Not being able to
find ideas

...and expressing the idea
is a frustrating thing.”



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